

Insert

1)

Activity - 3

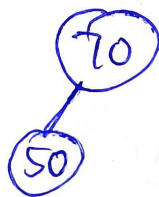
1) Construct an AVL tree

70, 50, 30, 90, 80, 130, 120 and apply the deletion operation to 120 and 30.

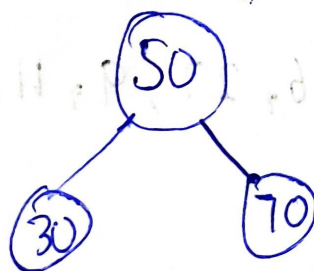
Insert 70



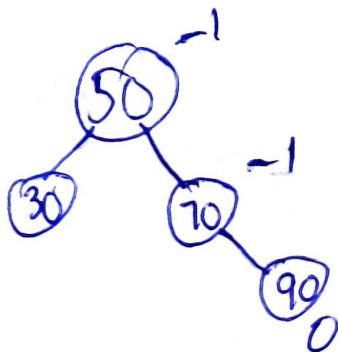
Insert 50



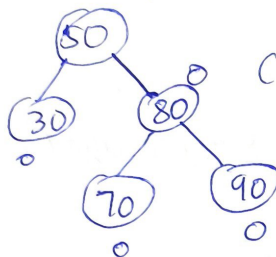
Insert 30



Insert 90

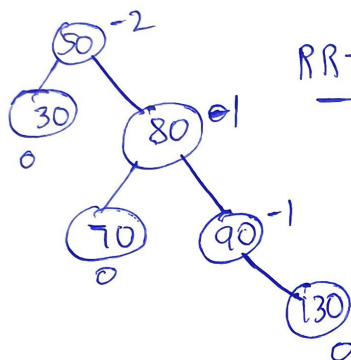


Insert 80

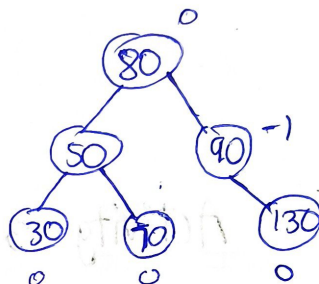


CR-L Rotation

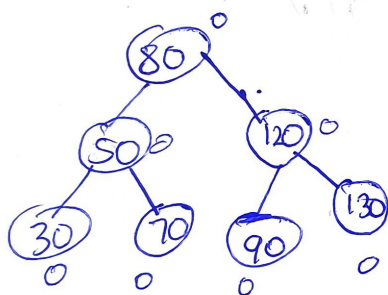
Insert 130



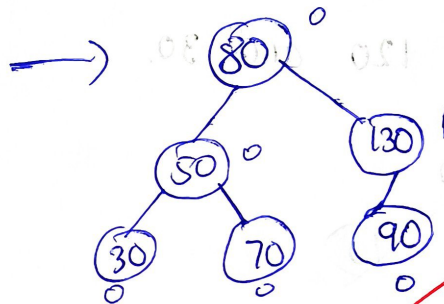
RR-Rotation



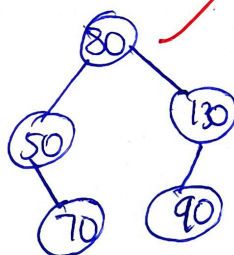
Insert 120



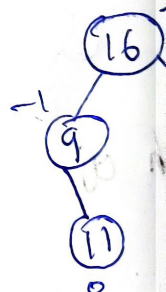
Delete 120



↓ Delete 30



Insert 11



2) AVL Tree - 16, 27, 9, 11, 36, 54, 81, 63, 72

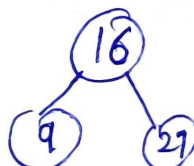
Insert 16



Insert 27



Insert 9



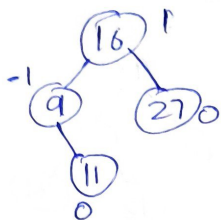
3) Create

11, 25,

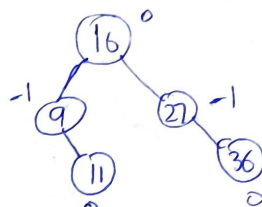
Insert

Insert

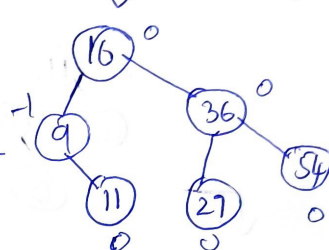
Insert 11



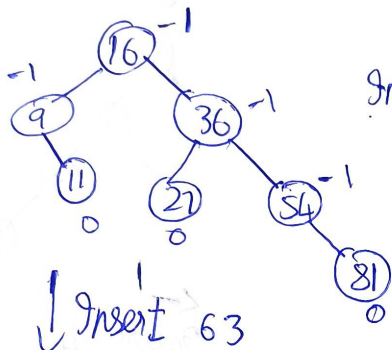
Insert 36



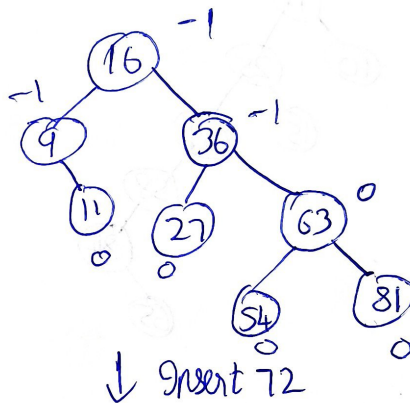
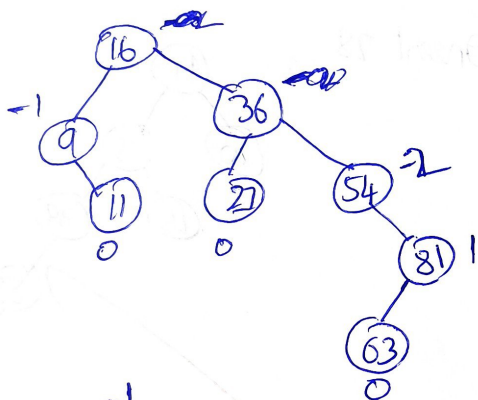
Insert 54



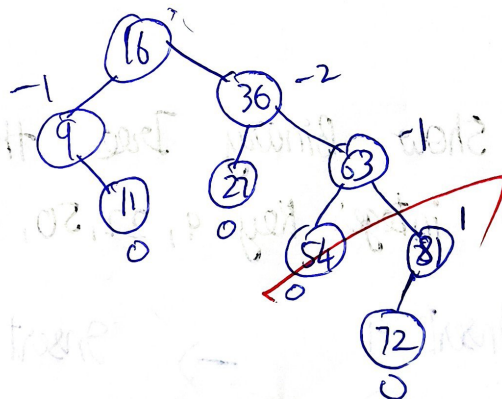
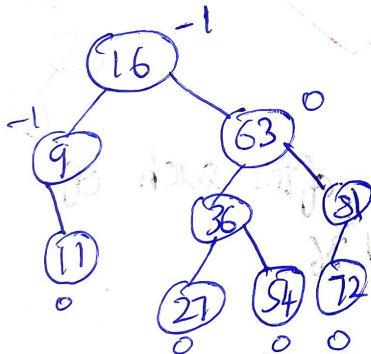
Insert 81



Insert 63



Insert 72



3) Create a Binary Search tree using following data
11, 25, 10, 15, 39, 58, 84, 65, 78.

Insert 11



Insert 25

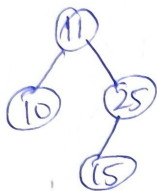


→

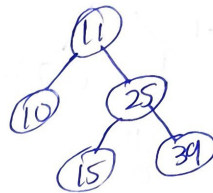
Insert 10



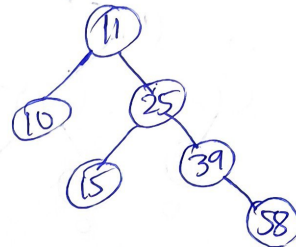
Insert 15



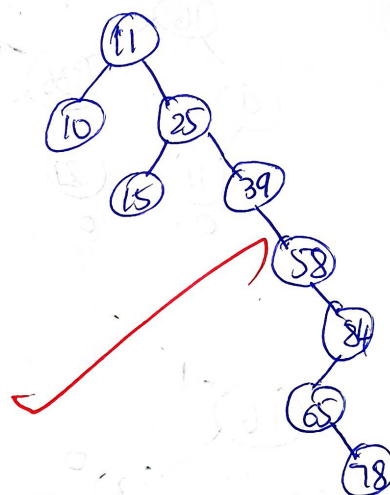
Insert 39



↓ Insert 58

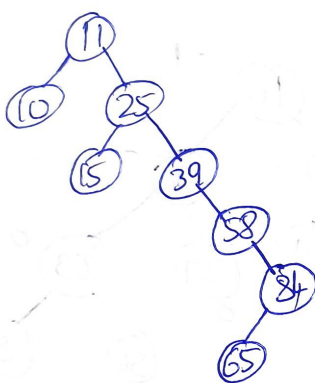


← Insert 84



Insert 78

Insert 65



4) Show Binary Tree that results after each of the integer Key 9, 27, 50, 15, 2, 21 and 36.

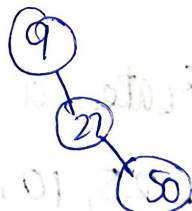
Insert 9



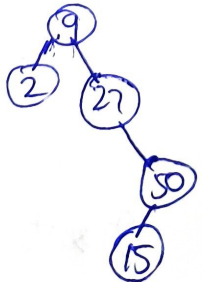
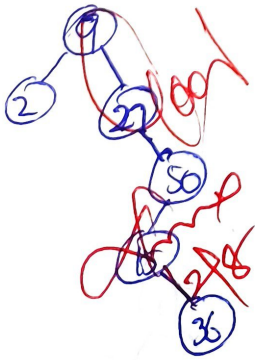
→ Insert 27



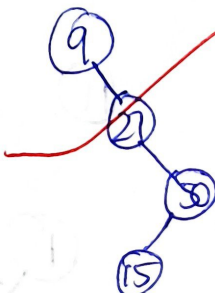
→ Insert 50



Insert 36



Insert 2



Insert 15